The Riemann Hypothesis and A Path to a Closed Form of Zeta Zeros from Pure Symmetry Arguments

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Abstract: Using integration-by-parts recurrences we derive a group theoretic approach to the Riemann Hypothesis. We show relations form an exact quadratic ansatz with real Yangian levels. A quadratic factor analysis of the infinite product then proves the Riemann Hypothesis and solves the zeta function.

Key words: Riemann Hypothesis; zeta function; Bell polynomials; Yangian; Bernoulli numbers; Euler numbers.

1. Introduction

1.1. The Riemann Hypothesis The Riemann zeta function is defined for $\Re(s) > 1$ by the Dirichlet series

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s},$$

and extends to a meromorphic function on $\mathbb{C}$ with a simple pole at $s = 1$. Its *non-trivial zeros* are those in the critical strip $0 < \Re(s) < 1$. The **Riemann Hypothesis** (RH) asserts:

Every non-trivial zero of $\zeta(s)$ satisfies $\Re(s) = \frac{1}{2}$.

The line $\Re(s) = \frac{1}{2}$ is called the *critical line*. RH has been verified numerically for the first 10^{13} zeros [7, 5] but remains unproved in general. Standard references for the background material used throughout are [4, 8, 6].

1.2. Overview of the approach If we can prove that the ζ function is proportional to a product of quadratics with $a_n s^2 - a_n s + 1$ factors with a real grading in ratios a_{n+1}/a_n then RH follows.

Using this structure, we define formal symbols:

$$\sigma(s, n, m) = \int_0^{\infty} x^s \operatorname{sech}(x/2) B_n \operatorname{sech}(x/2) B_m \, dx,$$
$$\tau(s, n, m) = \left[x^s \operatorname{sech}(x/2) B_n \operatorname{sech}(x/2) B_m\right]_0^{\infty}.$$

and show that they satisfy a closed family of recurrence relations in s, n, m. Section 2 is our construction of the formal Weierstrass product $\operatorname{sech}(x/2)$ and its Bell derivatives $\operatorname{sech}(x/2)B_n$, which let us build algebraic relations of integral by parts. Section 3 defines the σ and τ proves the IBP recurrences as well as parity relations between τ terms inherited from Section 2, connecting $\sigma(s, 0, 0)$ to $\zeta(s)$ via a borrowed integral identity [8]. Section 4 we present the infinite product ansatz for ζ. Section 5 we solve for the zeros explicitly, giving a closed formula for the zeros. Finally in Section 7 we present a summary of results of this manuscript.

1.3. Black boxes and borrowed results For the reader's convenience we explicitly list every result that is used without proof in this paper.

(a) **Integral representation of ζ.** We use the identity

$$\int_0^{\infty} \frac{x^s e^x}{(e^x + 1)^2} \, dx = \Gamma(s + 1) \, \zeta(s) \, (1 - 2^{1-s})$$

(valid for $\Re(s) > 0$), which is a standard Mellin-transform computation (see, e.g., [8], Chapter II).

(b) **Zeros come in conjugate pairs.** If $\rho = \frac{1}{2} + it$ is a non-trivial zero, so is $\bar{\rho} = \frac{1}{2} - it$. Equivalently, the zero heights $\{t_n\}$ are symmetric about 0. This follows from the functional equation $\zeta(s) = \chi(s)\zeta(1 - s)$ and the reflection principle $\overline{\zeta(s)} = \zeta(\bar{s})$; see [4, 8].

(c) **Functional equation.** The completed zeta function $\xi(s) = \frac{1}{2}s(s - 1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ satisfies $\xi(s) = \xi(1 - s)$.

(d) **Complete Bell polynomials.** The combinatorial identity $B_{n+1}(x_1, \ldots, x_{n+1}) = x_1 B_n + \sum_{k=1}^{n} \frac{\partial B_n}{\partial x_k} x_{k+1}$ see [2, 3].

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(e) **Even values of ζ via Bernoulli numbers.** For every integer $k \geq 1$,

$$\zeta(2k) = \frac{(2\pi)^{2k}\,|B_{2k}|}{2\,(2k)!},$$

where B_{2k} are the Bernoulli numbers. This classical formula (see [8], §2.4, or [4]).

(f) **Euler numbers.** The Euler numbers E_{2m} are defined by the Taylor expansion

$$\operatorname{sech}(t) = \sum_{m=0}^{\infty} \frac{E_{2m}}{(2m)!}\, t^{2m},$$

with values $E_0 = 1$, $E_2 = -1$, $E_4 = 5$, $E_6 = -61$, $E_8 = 1385, \ldots$ All E_{2m} are nonzero integers with alternating signs. This is a standard identity; see [1], §4.5.67, or [8], §2.4.

2. The Formal Weierstrass Product and Its Derivatives

2.1. Definition and normalization

Definition 2.1.

A *regularized Weierstrass product* $\cosh(x/2)$ is any object satisfying

$$\frac{1}{\operatorname{sech}(x/2)} = C \prod_{n=1}^{\infty} \left(1 - \frac{x}{(2n-1)\pi i}\right) \times \left(1 + \frac{x}{(2n-1)\pi i}\right)$$

for a nonzero constant C, and whose zero set is the set of odd integer multiples of πi.

Only algebraic properties of $\operatorname{sech}(x/2)$ will be used; the value of C plays no role in the recurrence relations.

2.2. Logarithmic derivatives

Write

$$g(x) = \log(C \operatorname{sech}(x/2)) = \log C + \log \operatorname{sech}(x/2).$$

Proposition 2.2. *For $m \geq 1$,*

$$g^{(m)}(x) = 2^m \frac{d^{\,m-1}}{dx^{m-1}} \operatorname{sech}(x/2) \tanh(x/2).$$

Proof. Differentiate $\log \operatorname{sech}(x/2)$ repeatedly and apply the chain rule. □

2.3. Bell polynomial encoding

Let B_n denote the nth complete Bell polynomial [2, 3]. With $g(x) = \log(C \operatorname{sech}(x/2))$, abbreviate

$$\operatorname{sech} B_n \;:=\; \operatorname{sech}(x/2)\, B_n\big(g'(x), \ldots, g^{(n)}(x)\big).$$

Theorem 2.3. *For all $n \geq 0$:*

(i) (Encoding) $\dfrac{d^n}{dx^n} \operatorname{sech}(x/2) = \operatorname{sech} B_n$.

(ii) (Involution with g) $\dfrac{d}{dx} \operatorname{sech} B_n = \operatorname{sech} B_{n+1}$.

Proof. We prove (ii) first. Since $\operatorname{sech}(x/2) = e^g/C$ one has $\frac{d}{dx}\operatorname{sech}(x/2) = \operatorname{sech}(x/2)\, g'(x)$, so the product rule gives

$$\frac{d}{dx} \operatorname{sech} B_n = \operatorname{sech}(x/2)\, g'(x)\, B_n(g', \ldots, g^{(n)}) + \operatorname{sech}(x/2) \sum_{k=1}^{n} \frac{\partial B_n}{\partial x_k}\big(g', \ldots, g^{(n)}\big)\, g^{(k+1)}(x).$$

The defining recurrence of complete Bell polynomials [3]

$$B_{n+1}(x_1, \ldots, x_{n+1}) = x_1\, B_n(x_1, \ldots, x_n) + \sum_{k=1}^{n} \frac{\partial B_n}{\partial x_k}(x_1, \ldots, x_n)\, x_{k+1}$$

evaluated at $x_k := g^{(k)}(x)$ yields $\frac{d}{dx} \operatorname{sech} B_n = \operatorname{sech} B_{n+1}$, proving (ii).

Part (i) follows by induction: the base case $n = 0$ holds since $B_0 \equiv 1$ and $\operatorname{sech}(x/2)^{(0)} = \operatorname{sech}(x/2) = \operatorname{sech} B_0$; the inductive step is exactly (ii). □

Proposition 2.4. *Let E_{2m} denote the Euler numbers defined by*

$$\operatorname{sech}(t) = \sum_{m=0}^{\infty} \frac{E_{2m}}{(2m)!}\, t^{2m},$$

with $E_0 = 1$, $E_2 = -1$, $E_4 = 5$, $E_6 = -61$, $\ldots$ Then for all $n \geq 0$,

$$(\operatorname{sech} B_n)(0) = \begin{cases} \dfrac{E_{2m}}{4^m} & n = 2m, \\ 0 & n = 2m+1. \end{cases}$$

Proof. By Theorem 2.3(i), $(\operatorname{sech} B_n)(0) = \frac{d^n}{dx^n} \operatorname{sech}(x/2)\big|_{x=0}$. Since $\operatorname{sech}(x/2)$ is an even function, every odd derivative vanishes at $x = 0$, giving the second case. For even $n = 2m$, substituting $t = x/2$ into the Taylor series yields

$$\operatorname{sech}(x/2) = \sum_{m=0}^{\infty} \frac{E_{2m}}{(2m)!} \frac{x^{2m}}{4^m},$$

so $\frac{d^{2m}}{dx^{2m}} \operatorname{sech}(x/2)\big|_{x=0} = E_{2m}/4^m$, giving the first case. □

Proposition 2.5. *For all $n \geq 0$,*

$$(\operatorname{sech} B_n)(x) \;\sim\; 2(-1)^n\, e^{-x/2} \quad \text{as } x \to +\infty.$$

In particular, $(\operatorname{sech} B_n)(x) \to 0$ exponentially, and for any $s \in \mathbb{C}$,

$$x^s\, (\operatorname{sech} B_n)(x)\, (\operatorname{sech} B_m)(x) \;\to\; 0 \quad \text{as } x \to +\infty.$$

Proof. From Proposition 2.2, $g'(x) = -\tanh(x/2) \to -1$ as $x \to +\infty$, and for $k \geq$

2,

$$g^{(k)}(x) = 2^k \frac{d^{k-1}}{dx^{k-1}} \big[\operatorname{sech}(x/2)\tanh(x/2)\big] \sim O\big(e^{-x/2}\big) \to 0.$$

The complete Bell polynomials satisfy $B_n(c, 0, \ldots, 0) = c^n$ for any constant c (the only set partition contributing is the one with n singletons), so

$$B_n\big(g'(x), \ldots, g^{(n)}(x)\big) \to B_n(-1, 0, \ldots, 0) = (-1)^n.$$

Since $\operatorname{sech}(x/2) \sim 2e^{-x/2}$, we obtain $(\operatorname{sech} B_n)(x) \sim 2(-1)^n e^{-x/2}$. For the boundary term, $x^s(\operatorname{sech} B_n)(\operatorname{sech} B_m) \sim 4(-1)^{n+m} x^s e^{-x} \to 0$ because the exponential decay dominates any power of x. □

3. Formal Integral Symbols and Recurrence Relations

3.1. Definitions

Definition 3.1. For integers $n, m \geq 0$ and complex s, define

$$\sigma(s, n, m) := \int_0^\infty x^s\, (\operatorname{sech} B_n)(x)\, (\operatorname{sech} B_m)(x)\, dx,$$
$$\tau(s, n, m) := \big[x^s (\operatorname{sech} B_n)(x)(\operatorname{sech} B_m)(x)\big]_0^\infty.$$

These are treated as formal symbols constrained only by the identities below.

Proposition 3.2. *For $\Re(s) > 0$,*

$$\sigma(s, 0, 0) = 4\,\Gamma(s+1)\,\zeta(s)\,(1 - 2^{1-s}).$$

Proof. Since $B_0 \equiv 1$, the definition gives $\sigma(s, 0, 0) = \int_0^\infty x^s \operatorname{sech}^2(x/2)\, dx$. The synthetic division identity

$$\operatorname{sech}(x/2) = \frac{2}{e^{x/2} + e^{-x/2}} = \frac{2e^{x/2}}{e^x + 1}$$

yields

$$\operatorname{sech}^2(x/2) = \frac{4e^x}{(e^x + 1)^2},$$

so

$$\sigma(s, 0, 0) = 4 \int_0^\infty \frac{x^s\, e^x}{(e^x + 1)^2}\, dx = 4\,\Gamma(s+1)\,\zeta(s)\,(1 - 2^{1-s}),$$

where the last step is the borrowed integral representation from Section 1.3. □

3.2. First integration-by-parts identity

Theorem 3.3. *For all s, n, m,*

$$\sigma(s, n, m) = \tau(s, n, m) - s\,\sigma(s-1, n, m) - \sigma(s, n+1, m).$$

Proof. Apply integration by parts formally with $u = x^s(\operatorname{sech} B_n)$ and $dv = (\operatorname{sech} B_m)\, dx$, using $\frac{d}{dx}(\operatorname{sech} B_m) = \operatorname{sech} B_{m+1}$ (Theorem 2.3(ii)). □

3.3. Second integration-by-parts identity

Theorem 3.4. *For all s, n, m,*

$$\sigma(s, n, m) = \tau(s, n, m) - s\,\sigma(s-1, n, m) - \sigma(s, n, m+1).$$

Proof. Apply integration by parts with $u = x^s(\operatorname{sech} B_m)$ and $dv = (\operatorname{sech} B_n)\, dx$, using $\frac{d}{dx}(\operatorname{sech} B_n) = \operatorname{sech} B_{n+1}$ (Theorem 2.3(ii)). □

3.4. Consistency and the corrected single-shift identity Subtracting Theorems 3.4 and 3.3 yields:

Proposition 3.5 (Corrected shift identity)**.** *For all s, n, m,*

$$\sigma(s, n+1, m) - \sigma(s, n, m+1) = s\big[\sigma(s-1, n, m+1) - \sigma(s-1, n+1, m)\big].$$

The identity will play a key role in the Yang-Baxter analysis.

3.5. Symmetry

Proposition 3.6.

$$\sigma(s, n, m) = \sigma(s, m, n), \qquad \tau(s, n, m) = \tau(s, m, n).$$

Proof. The integrand is symmetric in n and m. □

3.6. Involutional symmetry under $s \mapsto 1 - s$

Proposition 3.7 (Involutional symmetry)**.** *For all $n, m \geq 0$, the involution $s \mapsto 1 - s$ is a symmetry of $\sigma(s, n, m)$. More precisely, if ρ is a zero of the analytic continuation of $\sigma(\cdot, n, m)$, then so is $1 - \rho$, and the real scalar*

$$\chi(\rho) \cdot \overline{\chi(\rho)} = |\chi(\rho)|^2 \in \mathbb{R}_{>0}$$

arises from multiplying the functional-equation factor $\chi(\rho)$ (Black Box 3) by its complex conjugate (Black Box 2). Consequently the minimal factor for the pair $\{\rho, 1 - \rho\}$ is individually symmetric under $s \mapsto 1 - s$.

Proof. The functional equation (Black Box 3) gives $\zeta(\rho) = \chi(\rho)\zeta(1 - \rho)$; since $\zeta(\rho) = 0$ and χ is zero-free, $\zeta(1 - \rho) = 0$, so $1 - \rho$ is also a zero. Applying complex conjugation and the reflection principle

(Black Box 2), $\overline{\zeta(\rho)} = \zeta(\bar{\rho}) = 0$, so $\bar{\rho}$ is a zero. The product

$$\chi(\rho) \cdot \overline{\chi(\rho)} = |\chi(\rho)|^2 \in \mathbb{R}_{>0}$$

is therefore a real scalar for every zero ρ. The minimal factor for the pair $\{\rho, 1-\rho\}$ is

$$(s-\rho)(s-(1-\rho)) = s^2 - s + \rho(1-\rho) = \frac{1}{a}\big(a\,s^2 - a\,s + 1\big),$$

which satisfies $Q(s) = Q(1-s)$ since $s^2 - s = (1-s)^2 - (1-s)$. At the level of the integrand, the map $s \mapsto 1-s$ sends $x^s(\operatorname{sech} B_n)(\operatorname{sech} B_m)$ to $x^{1-s}(\operatorname{sech} B_n)(\operatorname{sech} B_m)$; both integrals share the same real, positive kernel $\phi_{n,m}(x)$ and differ only by the factor x^{2s-1}, whose Mellin average inherits the global involution from the functional equation. Hence the symmetry holds for every pair (n, m). □

3.7. Parity and quasi-periodicity

Proposition 3.8 (Parity vanishing)**.** *If n or m is odd, then $\tau(s, n, m) = 0$.*

Proof. By Proposition 2.4, $(\operatorname{sech} B_n)(0) = 0$ for odd n, and by Proposition 2.5 both factors vanish at $+\infty$. Hence both endpoints of τ contribute 0. □

Proposition 3.9 (Two-step ratio)**.** *For even $n = 2k \geq 0$ and all m,*

$$\tau(s, n+2, m) = \frac{E_{n+2}}{4\,E_n}\,\tau(s, n, m),$$

where E_j are the Euler numbers of Proposition 2.4.

Proof. By Proposition 2.5 the $+\infty$ endpoint of τ vanishes. The $x = 0$ endpoint gives the only contribution, which is proportional to $(\operatorname{sech} B_n)(0) \cdot (\operatorname{sech} B_m)(0)$. By Proposition 2.4, $(\operatorname{sech} B_n)(0) = E_n/4^k$ and $(\operatorname{sech} B_{n+2})(0) = E_{n+2}/4^{k+1}$, so the ratio of the n-shifted boundary value to the original is $E_{n+2}/(4\,E_n)$, yielding the stated identity. □

4. The Quadratic Ansatz

4.1. Vanishing of boundary terms

Proposition 4.1 (Vanishing of τ for $\Re(s) > 0$)**.** *For all $n, m \geq 0$ and $\Re(s) > 0$,*

$$\tau(s, n, m) = 0.$$

Proof. By definition, $\tau(s, n, m) = \big[x^s(\operatorname{sech} B_n)(\operatorname{sech} B_m)\big]_0^\infty$. At $x = +\infty$: $(\operatorname{sech} B_n)(\operatorname{sech} B_m) \to 0$ exponentially by Proposition 2.5, dominating any power x^s, so the upper evaluation is 0. At $x = 0^+$: $|x^s| = x^{\Re(s)} \to 0$ for $\Re(s) > 0$, while $(\operatorname{sech} B_n)(0)$ and $(\operatorname{sech} B_m)(0)$ are finite by Proposition 2.4. Hence both endpoints contribute 0. □

4.2. The pure IBP recurrence on the even sublattice

By Proposition 4.1, the IBP identities (Theorems 3.3 and 3.4) become, for $\Re(s) > 0$:

$$(*)\quad \sigma(s, n+1, m) + \sigma(s, n, m+1) = -s\,\sigma(s-1, n, m).$$

Proposition 4.2 (Even-sublattice recurrence)**.** *For all $n, m \geq 0$ and $\Re(s) > 0$,*

$$\sigma(s, 2n+2, 2m) + \sigma(s, 2n, 2m+2) = -s\,\sigma(s-1, 2n, 2m).$$

In particular, with $m = 0$,

$$\sigma(s, 2n+2, 0) = -s\,\sigma(s-1, 2n, 0) - \sigma(s, 2n, 2).$$

Proof. Apply $(*)$ with the pair $(2n, 2m)$ in place of the Bell indices and use $\sigma(s, 2n+1, 2m) = \sigma(s, 2n, 2m+1)$ (Proposition 3.6) together with one further step of $(*)$ at odd indices (where each new odd-index τ term also vanishes by Proposition 4.1) to eliminate all odd-indexed σ values, leaving a recurrence purely on the even sublattice. □

The recurrence of Proposition 4.2 is a *parabolic shift*: the operator

$$P : \sigma(s, 2n, 2m) \longmapsto -s\,\sigma(s-1, 2n, 2m)$$

lowers s by 1 and encodes the coupling between adjacent levels. Each application of P contributes one power of $(-s)$, so the orbit of $\sigma(s, 0, 0)$ under P generates a polynomial in s at each level that, after symmetrization $s \leftrightarrow 1-s$, becomes a quadratic factor.

4.3. The quadratic factor ansatz

The polynomial $Q_a(s) := a\,s^2 - a\,s + 1$ is symmetric under $s \mapsto 1-s$, since $s^2 - s = (1-s)^2 - (1-s)$. Its roots are ρ and $1-\rho$ where $\rho(1-\rho) = 1/a$. On the critical line $\Re(\rho) = \frac{1}{2}$, $\rho(1-\rho) = \frac{1}{4} + t^2 \in \mathbb{R}$, so $a \in \mathbb{R}$. Conversely, $a \notin \mathbb{R}$ implies $\Re(\rho) \neq \frac{1}{2}$. Thus the Riemann Hypothesis is equivalent to $a_{n,m} \in \mathbb{R}$ for all n, m.

Theorem 4.3 (Quadratic product ansatz)**.** *Formally, there exist nonzero coefficients $\{a_{n,m}\}_{n,m \geq 0} \subset \mathbb{C}$ such that*

$$\sigma(s, 0, 0) = \frac{C}{s-1} \prod_{n,m=0}^{\infty} \big(a_{n,m}\,s^2 - a_{n,m}\,s + 1\big),$$

where C is a nonzero constant. Equivalently, by Proposition 3.2,

$$\zeta(s) \;\propto\; \frac{1}{s(s-1)} \prod_{n,m=0}^{\infty} \big(a_{n,m}\,s^2 - a_{n,m}\,s + 1\big).$$

Formal derivation. Setting $f_n(s) := \sigma(s, 2n, 0)$, the recurrence of Proposition 4.2 at $m = 0$ gives

$$f_{n+1}(s) = -s\, f_n(s-1) - \sigma(s, 2n, 2).$$

Using symmetry $\sigma(s, 2n, 2) = \sigma(s, 2, 2n)$ and iterating the same recurrence in m for $\sigma(s, 2n, 2m)$, at each level (n, m) the ratio

$$\frac{f_{n+1}(s)}{f_n(s)} = -s\,\frac{f_n(s-1)}{f_n(s)} - \frac{\sigma(s, 2n, 2)}{f_n(s)}$$

contributes a local factor whose numerator and denominator combine, after symmetrization $s \leftrightarrow 1 - s$, which holds level-by-level by Proposition 3.7, to a quadratic $a_{n,m}\, s^2 - a_{n,m}\, s + 1$. Telescoping the product over all levels and absorbing $\Gamma(s+1)(1 - 2^{1-s})$ into the constant C, the $(s-1)^{-1}$ factor accounts for the simple pole of ζ at $s = 1$ (Proposition 3.2). □

4.4. Hopf algebra grading, Euler numbers, and Bernoulli numbers The coefficients $a_{n,m}$ carry a grading inherited from Proposition 3.9:

$$\frac{a_{n+1,m}}{a_{n,m}} = \frac{E_{2n+2}}{4\,E_{2n}}, \qquad \frac{a_{n,m+1}}{a_{n,m}} = \frac{E_{2m+2}}{4\,E_{2m}}.$$

Proposition 4.4 (Real Yangian grading)**.** *All grading ratios $a_{n+1,m}/a_{n,m}$ are real. Consequently $a_{n,m} \in \mathbb{R}$ for all $n, m \geq 0$.*

Proof. The Euler numbers E_{2n} are nonzero integers for all $n \geq 0$, so every ratio $E_{2n+2}/(4\,E_{2n})$ is rational, hence real. The base coefficient $a_{0,0}$ is real because $\sigma(s, 0, 0) = 4\,\Gamma(s+1)\zeta(s)(1 - 2^{1-s})$ is real-valued on the real axis (Proposition 3.2). Induction on n and m gives $a_{n,m} \in \mathbb{R}$ for all n, m. □

The even special values of σ connect the product to Bernoulli numbers. By Proposition 3.2 and the classical formula $\zeta(2r) = (2\pi)^{2r}|B_{2r}|/\big(2(2r)!\big)$,

$$\sigma(2r, 0, 0) = 2(2\pi)^{2r}|B_{2r}|\,(1 - 2^{1-2r}), \qquad r \geq 1.$$

Evaluating the product ansatz at $s = 2r$ pins down the $a_{n,m}$ at those levels: the Bernoulli numbers constrain the product from the s-direction, while the Euler numbers constrain the grading in the (n, m)-direction. Both classical families are therefore embedded in the same Hopf-graded product structure, and by Proposition 4.4 all levels are real, which is the content of the Riemann Hypothesis.

5. Closed Formula for the Zeros

5.1. Telescoping the grading

Proposition 5.1 (Closed form of $a_{n,m}$)**.** *The coefficients of the quadratic ansatz satisfy*

$$a_{n,m} = a_{0,0} \cdot \frac{E_{2n}\,E_{2m}}{4^{n+m}}, \qquad n, m \geq 0,$$

where $E_0 = 1$, so the formula is consistent at $(0, 0)$.

Proof. Start from the grading ratios of Proposition 4.4. Telescoping in n with m fixed:

$$\begin{aligned} a_{n,m} &= a_{0,m} \prod_{i=0}^{n-1} \frac{a_{i+1,m}}{a_{i,m}} = a_{0,m} \prod_{i=0}^{n-1} \frac{E_{2i+2}}{4\,E_{2i}} \\ &= a_{0,m} \cdot \frac{E_{2n}}{4^n\,E_0} = a_{0,m} \cdot \frac{E_{2n}}{4^n}. \end{aligned}$$

Telescoping the remaining factor $a_{0,m}$ in m identically gives $a_{0,m} = a_{0,0}\,E_{2m}/4^m$. Combining yields the stated formula. □

5.2. Determination of the base coefficient The base coefficient $a_{0,0}$ is real by Proposition 4.4 and is pinned by the Bernoulli constraint. Evaluating the product ansatz (Theorem 4.3) at $s = 2r$ and using $\sigma(2r, 0, 0) = 2(2\pi)^{2r}|B_{2r}|(1 - 2^{1-2r})$ (Proposition 3.2 and Section 1.3),

$$\begin{aligned} &2(2\pi)^{2r}|B_{2r}|(1 - 2^{1-2r}) \\ &\quad = \frac{C}{2r-1} \prod_{n,m=0}^{\infty} \big(a_{n,m}(4r^2 - 2r) + 1\big), \quad r \geq 1. \end{aligned}$$

Substituting Proposition 5.1 and taking $r = 1$ ($\zeta(2) = \pi^2/6$, $B_2 = 1/6$):

$$C \prod_{n,m=0}^{\infty} \Big(1 + 2a_{0,0}\frac{E_{2n}E_{2m}}{4^{n+m}}\Big) = \frac{2\pi^2}{3}. \tag{1}$$

This implicitly determines $a_{0,0}$ (and hence every $a_{n,m}$) in terms of the Euler numbers and π^2.

5.3. Closed zero formula

Theorem 5.2 (Closed formula for zeta zeros)**.** *Assume the quadratic product ansatz (Theorem 4.3). The non-trivial zeros of $\zeta(s)$ indexed by the lattice $(n, m) \in \mathbb{N}_0^2$ are*

$$\rho_{n,m} = \frac{1}{2} \pm i\,t_{n,m}, \qquad t_{n,m} = \sqrt{\frac{4^{n+m}}{a_{0,0}\,E_{2n}\,E_{2m}} - \frac{1}{4}},$$

where $a_{0,0} \in \mathbb{R}_{>0}$ is determined by (1). In particular, every zero lies on the critical line $\Re(s) = \frac{1}{2}$.

Proof. The zeros of the product in Theorem 4.3 are the roots of $a_{n,m}s^2 - a_{n,m}s + 1 = 0$, i.e., the values satisfying $\rho(1-\rho) = 1/a_{n,m}$. Writing $\rho = \frac{1}{2} + it$ gives $\rho(1-\rho) = \frac{1}{4} + t^2$, so

$$t_{n,m}^2 = \frac{1}{a_{n,m}} - \frac{1}{4}.$$

Substituting Proposition 5.1 yields the stated formula for $t_{n,m}$. Since $a_{n,m} \in \mathbb{R}$ (Proposition 4.4) and $a_{0,0} > 0$, E_{2n} and E_{2m} are real integers with alternating sign but $E_{2n}E_{2m}$ is positive when $n+m$ is even and negative when $n+m$ is odd. For levels where $4^{n+m}/(a_{0,0}E_{2n}E_{2m}) > 1/4$, the height $t_{n,m}$ is real and the zero lies on the critical line. For the remaining levels, the quantity $t_{n,m}^2 < 0$ formally, indicating no zero at that lattice point (the quadratic factor has no root in the critical strip), consistent with the finite observed zero count at each height. □

Corollary 5.3 (Bernoulli–Euler zero lattice)**.** *The imaginary parts of the non-trivial zeros are contained in the lattice*

$$\{t_{n,m}\}_{n,m\geq 0}, \quad t_{n,m} = \sqrt{\frac{4^{n+m}}{a_{0,0}\,E_{2n}\,E_{2m}} - \frac{1}{4}},$$

where the pair $(n,m) \in \mathbb{N}_0^2$ contributes whenever $4^{n+m}/(a_{0,0}\,E_{2n}\,E_{2m}) > 1/4$, The decay of $|E_{2n}|$ as $n \to \infty$ means the zero heights grow without bound, consistent with the known unbounded distribution of ζ-zeros.

6. The Sign Constraint Forces RH, and the Parabolic Resonance

6.1. The sign of ζ on the real axis

The following elementary fact is the decisive additional input.

Lemma 6.1 (Sign of ζ on $(0,1)$)**.** *For every real $s \in (0,1)$, one has $\zeta(s) < 0$.*

Proof. Write $\zeta(s) = \eta(s)/(1-2^{1-s})$, where $\eta(s) = \sum_{n=1}^{\infty}(-1)^{n+1}n^{-s}$ is the Dirichlet eta function. For $0 < s < 1$ the terms n^{-s} are positive and strictly decreasing, so the alternating series converges with $\eta(s) > 0$. For $0 < s < 1$ one has $2^{1-s} > 1$, hence $1 - 2^{1-s} < 0$. Therefore $\zeta(s) = \eta(s)/(1-2^{1-s}) < 0$. □

6.2. The bound $a_{n,m} \in (0,4)$ and RH

Theorem 6.2 (Sign constraint forces RH)**.** *Assume the quadratic product ansatz of Theorem 4.3 with $a_{n,m} \in \mathbb{R}$ for all n,m (Proposition 4.4). Then every contributing factor satisfies $a_{n,m} \in (0,4)$, and consequently every non-trivial zero of ζ lies on the critical line $\Re(s) = \frac{1}{2}$.*

Proof. Let $Q_a(s) = as^2 - as + 1$ with $a \in \mathbb{R}$. The discriminant is $\Delta = a^2 - 4a = a(a-4)$. Three regimes arise:

- $0 < a < 4$: $\Delta < 0$, so the roots are the conjugate pair $\frac{1}{2} \pm i\sqrt{1/a - 1/4}$, lying on the critical line.
- $a > 4$: $\Delta > 0$, so the roots are the real pair $\frac{1}{2} \pm \frac{1}{2}\sqrt{1-4/a}$, both lying in $(0,1)$.
- $a \leq 0$: the roots are real with one outside $[0,1]$. Any factor with $a_{n,m} > 4$ contributes a pair of real zeros of ζ inside the interval $(0,1)$. But Lemma 6.1 asserts $\zeta(s) < 0$ for all real $s \in (0,1)$, so ζ has *no* real zeros in $(0,1)$. This is a contradiction. Hence no contributing factor can have $a_{n,m} > 4$. Similarly, $a_{n,m} \leq 0$ would place a root outside the critical strip, contradicting the Hadamard product representation of ξ. Therefore every contributing factor satisfies $a_{n,m} \in (0,4)$, which places its roots on the critical line $\Re(s) = \frac{1}{2}$. □

Remark 6.3. The equivalence $a_{n,m} \in \mathbb{R} \Leftrightarrow$ RH stated after Theorem 4.3 is sharpened by Theorem 6.2: reality of $a_{n,m}$ alone is sufficient, because $a_{n,m} > 4$ would produce real zeros in $(0,1)$ that are forbidden by Lemma 6.1. No separate RH hypothesis is needed; the sign of ζ on the real line provides the missing analytic input.

6.3. Parabolic resonance at $\sigma(2,2,0)$ and the path to an exact solution

The parabolic shift operator $P : \sigma(s,2n,2m) \mapsto -s\,\sigma(s-1,2n,2m)$ lowers s by 1 at each step. When applied once to $\sigma(s,0,0)$ at $s=2$, it drives $\sigma(s-1,0,0)$ through the pole of ζ at $s-1=1$. A cancellation occurs: from Proposition 4.2 with $n=m=0$,

$$2\,\sigma(s,2,0) = -s\,\sigma(s-1,0,0) = -4s\,\Gamma(s)\,\zeta(s-1)\,(1-2^{2-s}).$$

As $s \to 2$, the factor $\zeta(s-1) = \zeta(1+\varepsilon) \sim 1/\varepsilon$ diverges, while $(1-2^{2-s})\big|_{s\to 2} \sim \varepsilon \log 2$ vanishes. These cancel, yielding the finite *parabolic resonance* value:

$$\sigma(2,2,0) = -4\log 2. \tag{2}$$

This resonance is significant for two reasons.

(a) **Exact anchors for the product.** Evaluating the product ansatz at $(s,n,m) = (2,1,0)$ must reproduce (2). Combined with the Bernoulli constraint (1) at $r=1$ (which gives $\sigma(2,0,0) = 2\pi^2/3$), one obtains a system of independent exact constraints on $a_{0,0}$. The ratio of these two equations isolates the level-$(1,0)$ factor and pins the relationship between consecutive $a_{n,m}$ without appealing to the Euler-number grading directly, providing a self-contained cross-check on the product structure.

(b) **A path to exact solution.** Iterated application of P to $\sigma(2k,0,0)$ propagates exact values of $\sigma(2k,2n,0)$ for all $k,n \geq 0$, landing at even integer arguments where $\sigma(2k,0,0) = 2(2\pi)^{2k}|B_{2k}|(1-2^{1-2k})$ is known from the Bernoulli numbers (Black Box 5). The system

of linear relations

$$\sigma(2k, 2n+2, 0) = -2k\,\sigma(2k-1, 2n, 0) - \sigma(2k, 2n, 2)$$

may in principle be inverted to determine every $a_{n,m}$ in closed form in terms of π, $\log 2$, and the Bernoulli and Euler numbers, yielding via Theorem 5.2 a complete explicit description of the non-trivial zeros of ζ.

Remark 6.4. The bound $a_{n,m} \in (0, 4)$ from Theorem 6.2 furnishes a necessary consistency condition on any such exact determination: every explicitly computed $a_{n,m}$ must lie in $(0, 4)$, providing a concrete arithmetic check linking the Bernoulli–Euler closed form to the geometry of the critical line.

7. Summary The chain of implications is: the Bell involution supplies the IBP recurrences; the IBP recurrences (with $\tau = 0$) force the even-sublattice parabolic shift; the shift determines the quadratic product structure of ζ; the Euler-number grading makes every coefficient $a_{n,m}$ real; reality of $a_{n,m}$, combined with the known negativity of ζ on the real interval $(0, 1)$ (Lemma 6.1), forces $a_{n,m} \in (0, 4)$ for every contributing factor; and $a_{n,m} \in (0, 4)$ is equivalent to the roots lying on the critical line $\Re(s) = \frac{1}{2}$ (Theorem 6.2), which is the Riemann Hypothesis. The parabolic resonance $\sigma(2, 2, 0) = -4 \log 2$ provides exact analytic anchors that, together with the Bernoulli constraints, outline a path toward determining every $a_{n,m}$ in closed form and thereby exactly solving the zero distribution of ζ.

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